

Simulation Note: 2-D Turbulence Runs for Synthetic PIV Data Generation

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1 The simulation

Each sample in the dataset comes from an independent direct simulation of **two-dimensional turbulence**, run with the pure-Julia `Oceananigans` solver. The configuration is identical across all runs; only the random seed changes from one simulation to the next.

- **Grid & domain:** 512×512 cells, doubly periodic, unit grid spacing ($\Delta x = 1$).
- **Numerics:** WENO advection (order 5); explicit scalar diffusivity $\nu = 1 \times 10^{-5}$; adaptive time step controlled by a CFL wizard ($\text{CFL} = 0.7$).
- **Initial condition:** a band of random spectral modes (up to wavenumber ~ 21) superposed with a coherent jet, giving a different, statistically independent flow for every seed.
- **Tracers:** passive particles are advected by the resolved velocity field and later rendered as the synthetic PIV particle images; the underlying velocity field is retained as the ground-truth label.

2 How long does each run last?

The save interval and total length are tied to each simulation's own peak speed $U_{\max}$, so the flow is sampled consistently regardless of how energetic it is. With $\Delta x = 1$ the code sets

$$\Delta t_{\text{save}} = \frac{5}{U_{\max}}, \quad t_{\text{stop}} = n_t \Delta t_{\text{save}}, \quad n_t = 40. \quad (1)$$

This yields the following per-run summary:

Saved frames	$n_t + 1 = 41$ snapshots (one every Δt_{save})
Simulation time	$t_{\text{stop}} = 40 \Delta t_{\text{save}} = 200/U_{\max}$ (code units)
Physical span (invariant)	fastest tracers travel ~ 200 cells ($\approx 40\%$ of the box)
Per-frame displacement	$s_{\max} = U_{\max} \Delta t_{\text{save}} \approx 5$ px <i>by construction</i>
Wall-clock	~ 6 minutes per run (one Unity CPU node)

Because $\Delta t_{\text{save}} = 5/U_{\max}$ is set from each run's initial speed, the *absolute* simulation time varies between runs, but the *displacement* budget is fixed: every frame interval advances the fastest particles by about five pixels.

3 Building the image pairs — where is the “first frame”?

Each sample provides three image pairs at nominal particle displacements of **10, 20, and 30 px**. All three pairs share the *same* first image A ; the second image B is taken dp frames later, where $dp = \lfloor \text{pix}/s_{\max} \rfloor$.

Importantly, frame A is not at the start of the run — it sits near the end. It is anchored so that the largest-displacement bin (30 px) places its second image B exactly on the final, most-developed frame:

$$i_A = n_{\text{frames}} - \max(dp) = 41 - 6 = \text{frame 35}. \quad (2)$$

Bin	First image A	dp	Second image B
10 px	frame 35	2	frame 37
20 px	frame 35	4	frame 39
30 px	frame 35	6	frame 41 (final frame)

Frame 35 corresponds to a simulation time $t_A \approx 170/U_{\max}$, i.e. roughly **85% of the way through the run**.

Why near the end? The initial condition is spectrally truncated (artificial, with no small-scale structure). Over the run the 2-D enstrophy cascade fills in the small scales, so the later frames hold the most physically developed turbulence — the best signal for training. Anchoring A near the end of the run captures that developed state.

Caveat (recorded per sample). Because $s_{\max}$ sits right at 5.0, the floor in $dp = \lfloor \text{pix}/s_{\max} \rfloor$ can shift i_A by ± 1 frame between runs. The exact anchor frame and frame count are therefore stored per sample (`frameA_index`, `nframes`) in the metadata, alongside the realized displacement for each bin.

4 The dataset

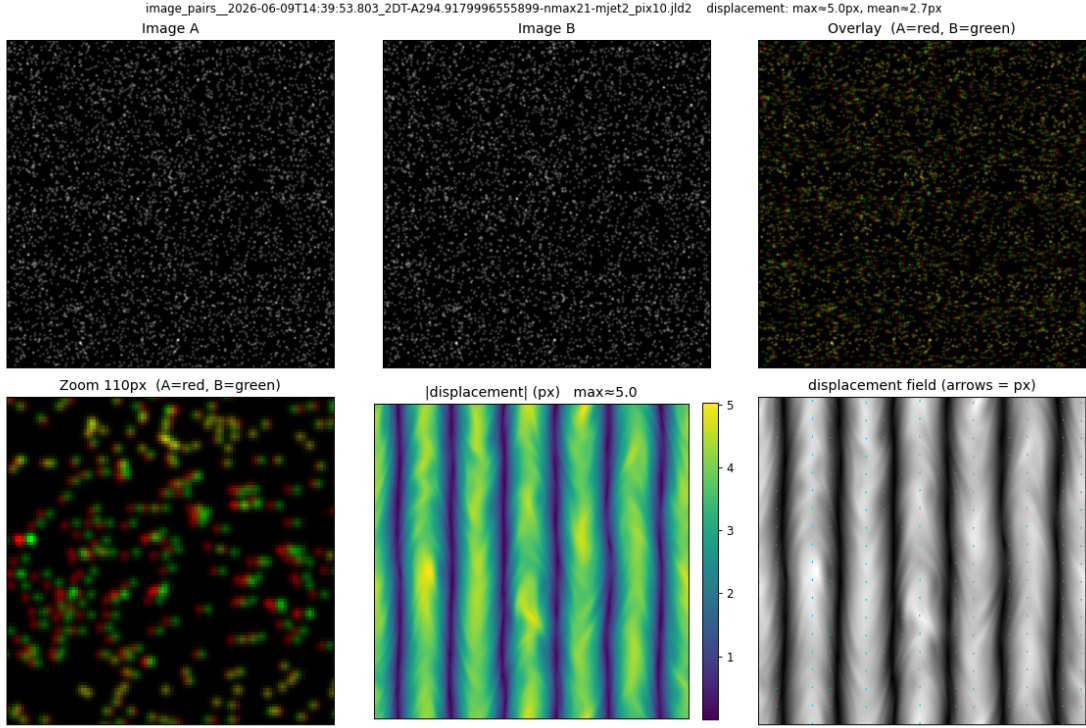
The production campaign generated **100,000 independent simulations** (one image-pair sample per simulation, per displacement bin) on the Unity HPC cluster:

Samples per bin (10/20/30 px)	100,000 each (300,000 pairs total)
Per-sample contents	images A, B + ground-truth velocity fields (u, v) at A and B
Metadata	one sidecar per sim $\rightarrow$ <code>manifest.parquet</code> ($10^5 \times 48$ columns)
Total size	~ 1.3 TB, archived to permanent <code>/project</code> storage

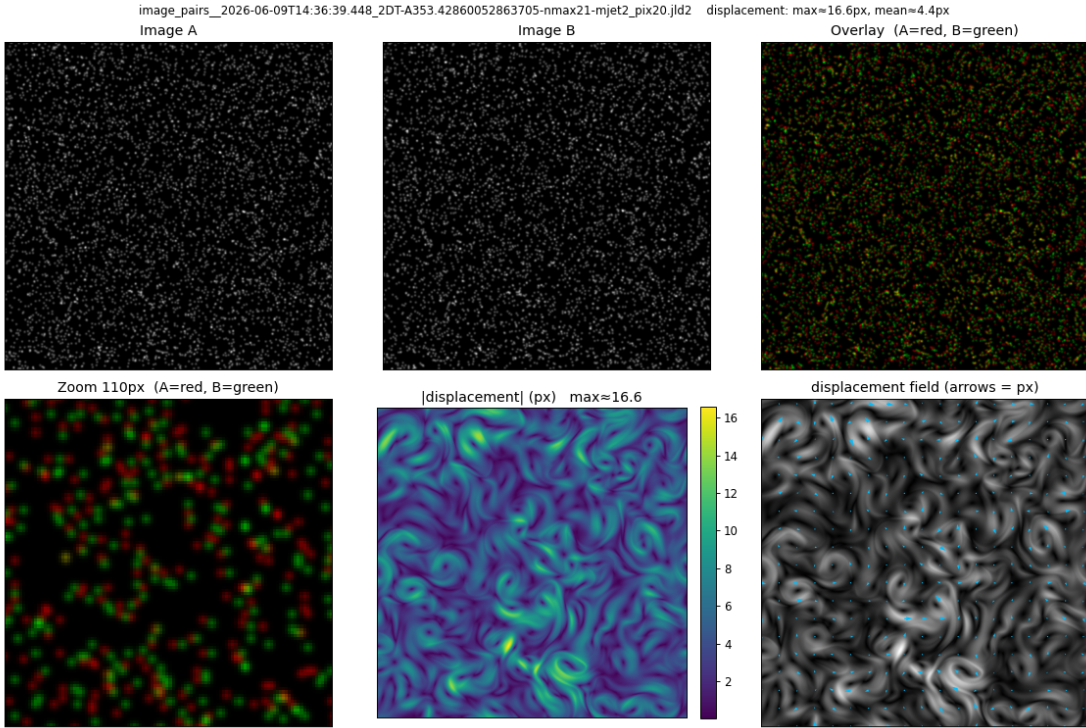
The manifest records, for every sample, the seed, all physics parameters, code git SHA, initial/final flow diagnostics, and the displacement bookkeeping — so any sample can be filtered, traced back, and exactly reproduced.

5 Sample image pairs

Figure 1 shows representative previews from the pipeline. Each preview is a six-panel view: *top row* — image A , image B , and an A/B overlay (red = A , green = B); *bottom row* — a zoomed overlay, the displacement magnitude, and the ground-truth displacement field. The slight red/green offset in the overlays is the prescribed particle displacement that the network learns to invert.



(a) 10 px displacement bin



(b) 20 px displacement bin

Figure 1: Representative synthetic PIV samples (different independent simulations). Particle images A and B are the network inputs; the displacement / velocity field is the ground-truth label.